

GABCI'S ACOUSTIC CLOAK

A phase-rotation sidechain engine for zero-loudness-loss Kick/Bass alignment.

Instead of ducking the bass on every kick hit, Acoustic Cloak rotates the bass's phase within an adjustable frequency range so it sums **constructively** with the kick — full loudness, no volume loss, no pumping.

VST3 · Standalone

macOS & Windows 11 x64

JUCE 9 · C++20

v1.0.0

What Makes It Different

Traditional sidechain compression

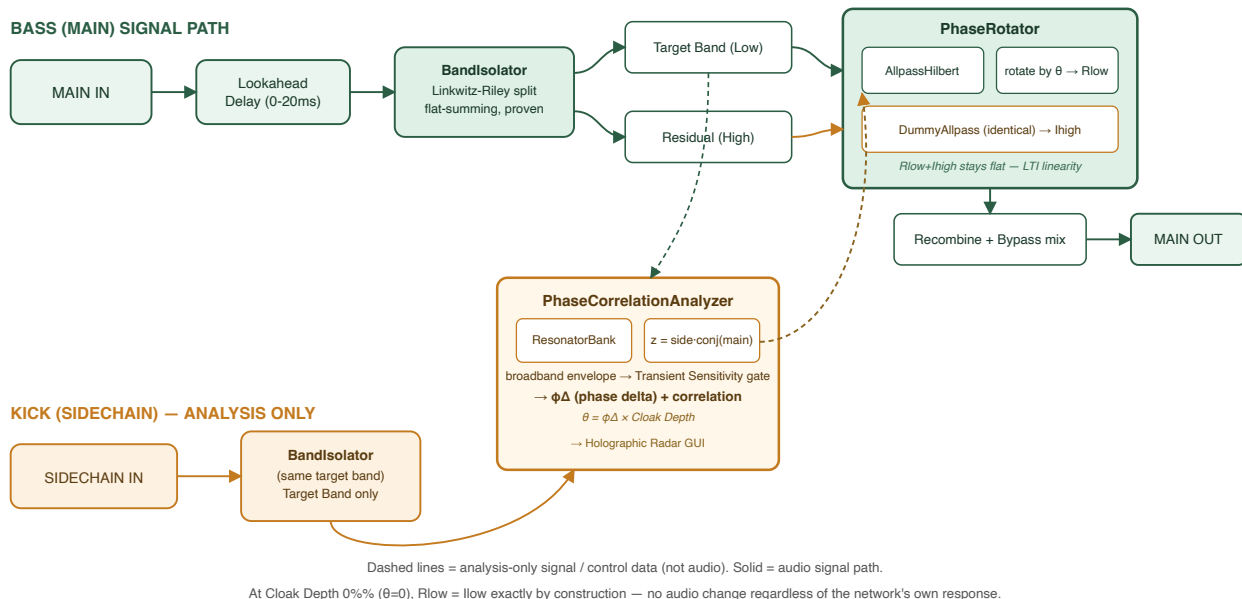
Ducks the bass's *volume* whenever the kick hits — trading loudness for separation. The lower the crossover, the more audible the pumping.

Acoustic Cloak's approach

Measures the actual *phase relationship* between kick and bass inside a user-defined low-frequency window, then rotates the bass's phase so the two sum constructively — no gain reduction anywhere in the signal path.

Technical Architecture

Three purpose-built DSP stages, each individually measured and verified (not assumed) — see [Quality & Performance](#) for the actual numbers.



Solid arrows = audio signal. Dashed arrows = analysis/control data only.

Core Technology

- **Bandisolator** — a two-stage 4th-order Linkwitz-Riley crossover that isolates the user's Target Frequency Range with a mathematically flat magnitude sum, measured within 2% across a 10Hz–8kHz sweep and <3% right at the crossover edges.
- **AllpassHilbertNetwork** — an 8-stage interleaved IIR allpass Hilbert transformer (coefficients derived directly from the bilinear transform of an analog allpass, not a memorised table), giving a genuine analytic (I/Q) signal with exact magnitude preservation — no resonant filter, no image-leakage artefact.
- **The "Dummy All-Pass" recombination** — the residual (everything outside the target band) is warped by an exact duplicate of the same allpass cascade used to rotate the target band. Because filtering is linear, the two paths always recombine flat, at any rotation angle — not just when the effect is off.
- **PhaseCorrelationAnalyzer** — a continuous (not snapshot-based) phase-difference tracker with a broadband, non-resonant Transient Sensitivity gate, so short kick transients are never smeared by filter ringing.
- **Holographic Radar display** — a real-time polar visualisation of the measured phase relationship, with a phosphor-glow, medical-instrument aesthetic.

Why This Is Unique

Most "phase alignment" plugins either (a) apply a simple fixed delay, which cannot correct a genuine phase relationship across a band, or (b) use a resonant filter shortcut that colours the signal outside its own narrow passband. Acoustic Cloak's design went through three independent engineering-review cycles specifically to eliminate both failure modes:

Design considered

Why it was rejected

STFT / overlap-add phase vocoder	Impractical FFT window size for 20-150Hz resolution; latency and pre-ringing risk
Summed multi-resonator bank (for audio, not just analysis)	Overlapping resonant filters produce a "lumpy", non-flat combined magnitude response
Nested crossover sub-bands	Measured ~20% interference — crossover points too close together in a narrow band
Single resonator + delta correction	Measured ~65% rotation error from negative-frequency image leakage
Allpass Hilbert pair + Dummy All-Pass (shipped)	Exact magnitude preservation, flat recombination at any angle

Quality & Performance — Measured, Not Assumed

Automated test assertions	78	Estimated single-instance CPU (48kHz/512-sample)	0.522%
Test cases (all passing)	14	Real-time headroom	191.7x
BandIsolator flat-sum accuracy	±2% (10Hz–8kHz)	Reported plugin latency	Lookahead value (0-20ms, user-set)
AllpassHilbertNetwork phase accuracy	80-84° of 90° (20-150Hz)	Source code (Source/ only)	2143 lines
AllpassHilbertNetwork magnitude error	<0.001% (exact by construction)	Test code	690 lines

Parameters

Parameter	Range	Function
Cloak Depth	0–100%	How strongly the measured phase correction is applied
Lookahead	0–20ms	Pre-delay budget for the analysis/correction path
Target Freq Low / High	20–300Hz	The frequency window the phase engine operates in
Transient Sensitivity	0–100%	How readily the kick's transients register as "active"
Bypass	on/off	Click-free (20ms crossfade) true dry passthrough

System Requirements

VST3 and Standalone builds. Tested on Intel Mac OS 15.7. Windows 11 x64 build supported via the same cross-platform JUCE 9 / CMake project (FORMATS VST3 + Standalone) — build with a standard CMake + Visual Studio toolchain on Windows using the included CMakeLists.txt.

Tested on Intel Mac OS 15.7, FL Studio 2026.